

Overview of Statistical Learning and Modeling – 36-600

Lab 14T: Data Analysis Workflow Review

Due Wednesday, December 3, 11:59pm

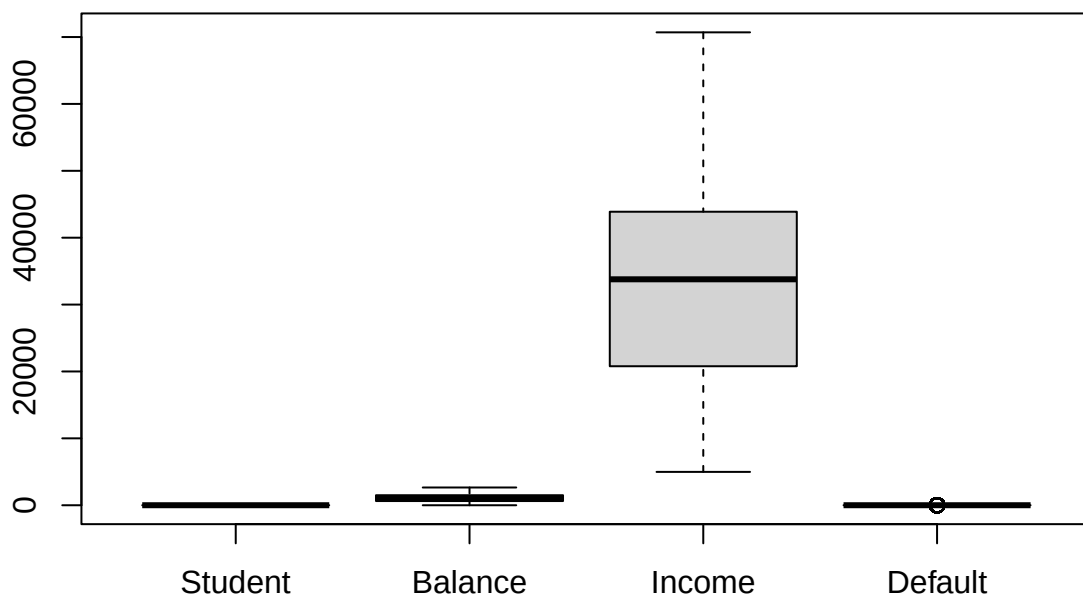
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This week's lab is very much DIY. There is a file called `creditDefault.csv` on the course Github page. Your boss wants to know if any of the variables included in this file, apart from `Default` itself, are important for predicting whether a person will default on their credit-card balance rather than paying it back.

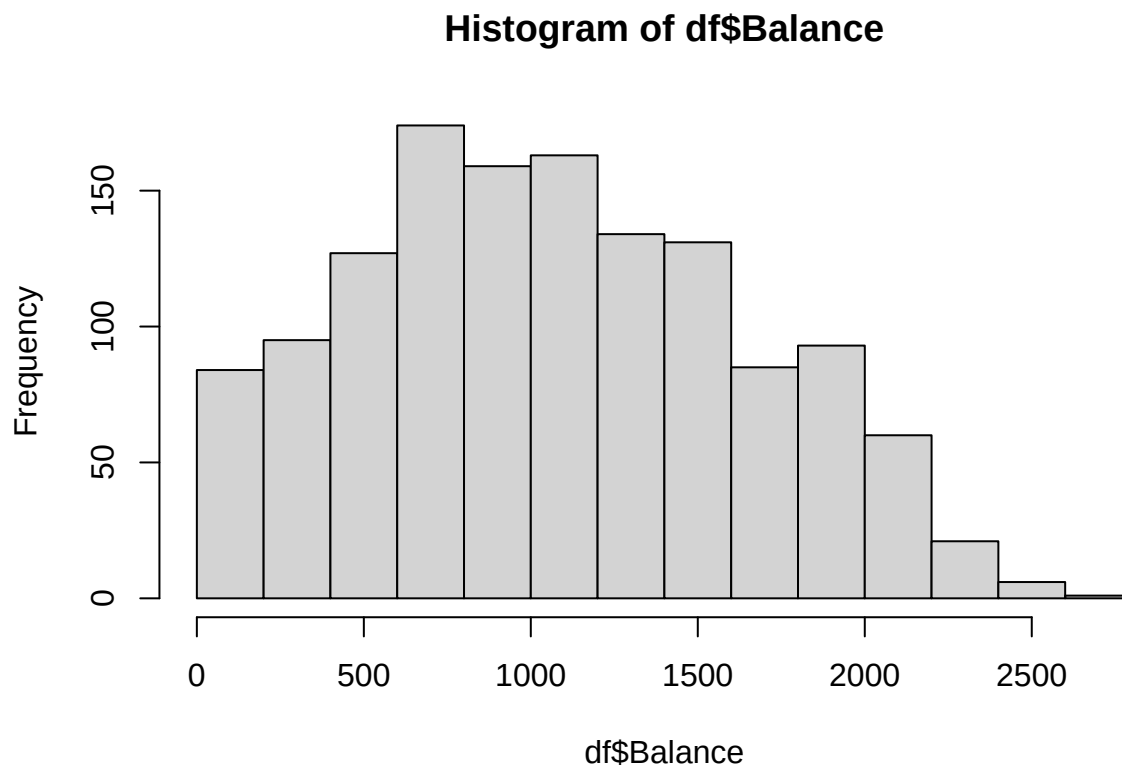
From just the information above, you should eventually construct an appropriate answer to the question. Note that regardless of whether you think a non-ML model is sufficient, you should always check to see if an ML model gives much better predictive results. Also note that your boss is not necessarily looking for plots here, but you'd always want to do *some* EDA, if for no other reason than to ensure you don't have missing data or outliers!

```
df <- read.csv("https://raw.githubusercontent.com/FSKoerner/F25-36600-data/refs/heads/main/creditDefault.csv")
```

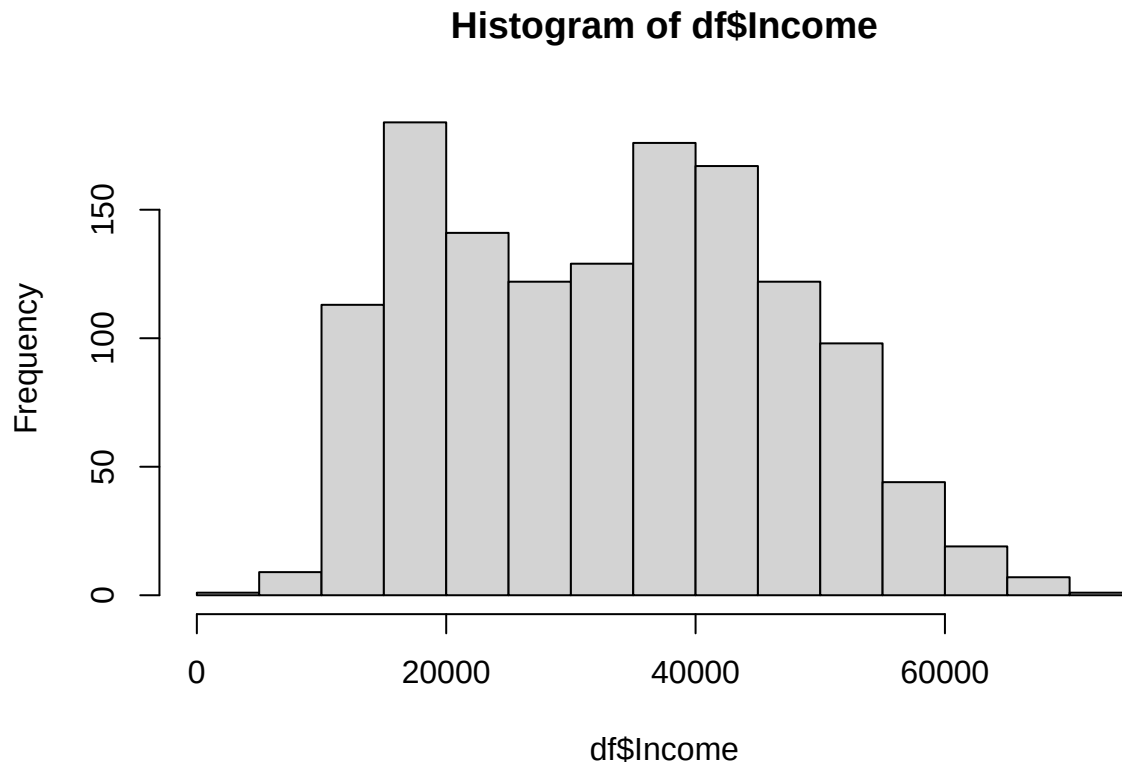
```
boxplot(df)
```



```
hist(df$Balance)
```



```
hist(df$Income)
```



```
set.seed(42)
s      <- sample(nrow(df), round(0.7 * nrow(df)))
df.train <- df[s, ]
df.test  <- df[-s, ]

logit.out <- glm(Default ~ ., data = df.train, family = binomial)
summary(logit.out)
```

```
##
## Call:
## glm(formula = Default ~ ., family = binomial, data = df.train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.027e+00  7.430e-01 -10.802  <2e-16 ***
## StudentYes   -8.514e-01  3.947e-01  -2.157   0.031 *
## Balance       5.338e-03  3.729e-04  14.316  <2e-16 ***
## Income        2.311e-06  1.341e-05   0.172   0.863
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1044.34  on 932  degrees of freedom
## Residual deviance:  486.35  on 929  degrees of freedom
```

```
## AIC: 494.35
##
## Number of Fisher Scoring iterations: 6

balance.grid <- seq(min(df.train$Balance), max(df.train$Balance), length = 300)

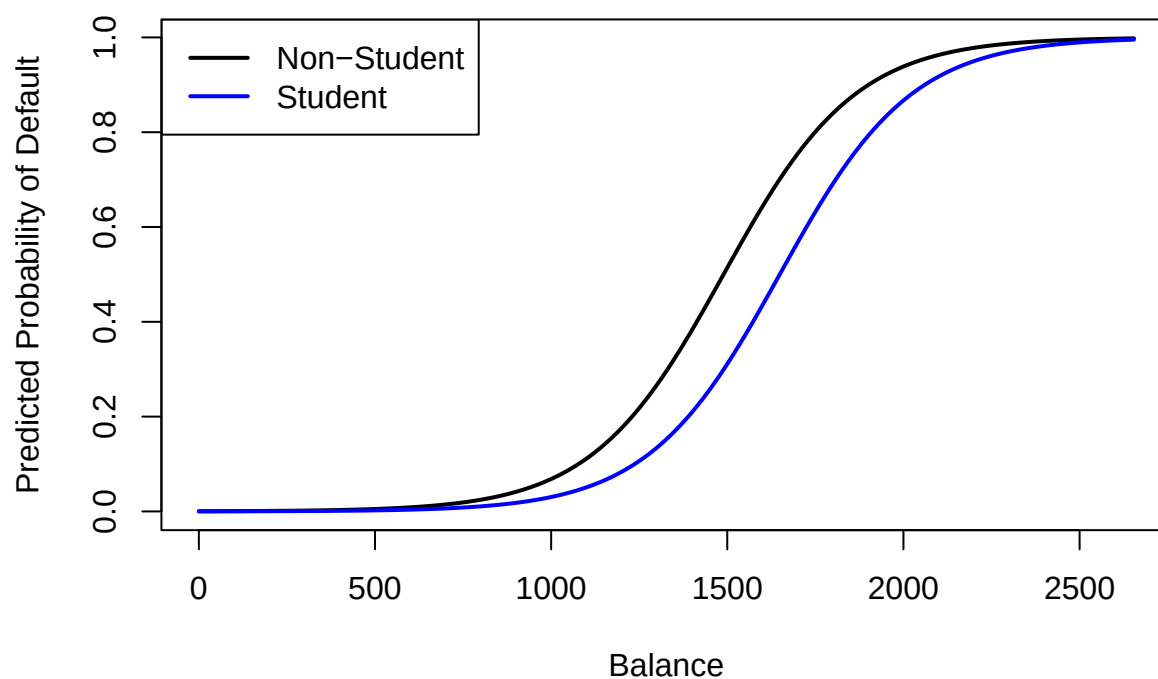
# Predict for non-student
pred.non <- predict(logit.out,
                    newdata = data.frame(
                      Student = "No",
                      Balance = balance.grid,
                      Income = mean(df.train$Income)
                    ),
                    type = "response")

# Predict for student
pred.stud <- predict(logit.out,
                    newdata = data.frame(
                      Student = "Yes",
                      Balance = balance.grid,
                      Income = mean(df.train$Income)
                    ),
                    type = "response")

# PLOT
plot(balance.grid, pred.non, type = "l", lwd = 2,
      xlab = "Balance", ylab = "Predicted Probability of Default",
      main = "Logistic Regression: Predicted Probability of Default")

lines(balance.grid, pred.stud, lwd = 2, col = "blue")
legend("topleft", legend = c("Non-Student", "Student"),
      col = c("black", "blue"), lwd = 2)
```

Logistic Regression: Predicted Probability of Default



Based on our analysis of the logistic regression model, we can see that Balance is significant in determining if a someone is likely to default.